

ROPE HAS A SPECTRAL BUDGET

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ABSTRACT

RoPE optimisation is usually framed through its base, spectral range, or phase transport, while geometric spacing silently fixes how a finite set of rotary pairs is allocated inside that range. Writing $x_k = -\log \omega_k = a + Rz_k$ exposes interior allocation z as a third finite-table design coordinate. Exact, phase-invariant sin/cos geometry shows that redundancy can collapse many nominal dimensions onto nearly the same two-dimensional positional basis. On a frozen billion-scale OLMo checkpoint, changing only z at matched support raises nine-task 16K RULER from 0.56% to 60.47%; a coarse label-free allocation reproduces this recovery without task labels or parameter updates. A paired three-seed fixed-support training intervention independently improves every tested OOD length in every seed, identifying z during co-adaptation. A stated convex surrogate then yields EVQ-COSH, a closed-form zero-learned-parameter construction unique for that surrogate. Across fully frozen, matched low-rank adaptation, and from-training interventions, the same coordinate remains consequential throughout the model lifecycle. A finite RoPE table is jointly determined by its sampled support and how its spectral budget is allocated.

1 INTRODUCTION

Most RoPE optimisation is described through a scalar base, a spectral range, or a transform of the phases realised at longer positions. That framing leaves a separate finite-table choice implicit: where a head places phase resolution among its exactly $K = d_{\text{rot}}/2$ rotary pairs. A scalar base determines the table only after geometric spacing has fixed this allocation. We separate the choices by writing

$$x_k = -\log \omega_k = a + Rz_k, \quad z_0 = 0, \quad z_{K-1} = 1, \quad (1)$$

where k indexes the fixed K rotary pairs, a sets support location, R its span, and z the normalised interior allocation; geometric RoPE fixes $z_k = k/(K-1)$. The third coordinate z decides which phase scales the fixed K samples occupy and where resolution is placed. We test it with support, K , and the RoPE operator fixed, then derive a closed-form construction.

This distinction matters because geometric spacing is highly redundant at the slow end. For $L=4096$, $b=5 \times 10^5$, and $K=64$, the 23 pairs with $\omega L \leq 1$ occupy 46 nominal dimensions yet have block-whitened Rényi-2 effective rank 2.00. These channels can still carry content, but as positional functions they converge toward the same two-dimensional subspace. A geometric table therefore spends more than one third of its rotary pairs on near-duplicate slow positional directions. This is the spectral-budget problem studied here.

Long-context methods can change realised phases, sampled support, or interior placement, often more than one at once. Learned-frequency methods also show that the geometric exponent sequence need not remain fixed. Section 5 classifies these interventions under the decomposition above. We isolate the missing variable by pinning both endpoints and the log-span, then moving only the interior allocation.

The strongest practical consequence appears after pretraining. At matched support, amplitude, checkpoint, rows, and decoder, changing only z raises OLMo nine-task 16K RULER from 0.56% to 60.47%, without parameter updates (§4.1). A separate fixed-support training control answers the causal question at model construction: across three paired 151.9M seeds, moving only the 30 interiors improves every tested OOD length in every seed (§2.1). Together these interventions show that z is neither a disguised base change nor a coordinate that disappears once weights have been trained.

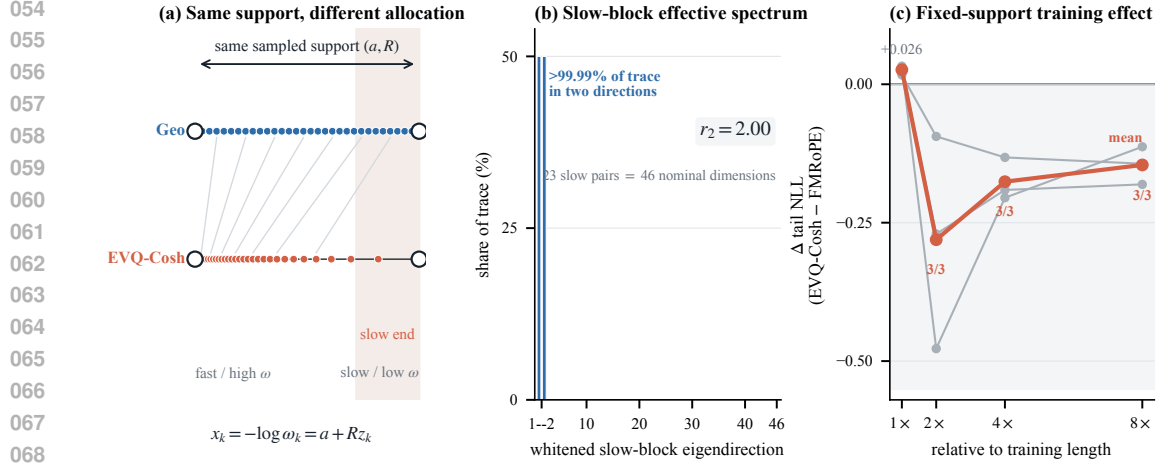


Figure 1: **Interior allocation is a causal coordinate of a finite rotary budget.** (a) Geo and anchored EVQ-COSH share $K=32$, operator, endpoints, and log-span; z changes the phase-scale density. (b) Exact geometry: for standard RoPE at $L=4096$, $b=5 \times 10^5$, and $K=64$, the 23 pairs with $\omega L \leq 1$ place more than 99.99% of their block-whitened trace in two eigendirections and have $r_2=2.00$ under the stated uniform separation prior. (c) Causal identification: moving only 30 interiors at fixed support improves every tested OOD length in all three seeds. The panels respectively define the coordinate, expose its geometric consequence, and isolate its training-time effect.

Exact Variational Quantization with a Cosh density (EVQ-COSH) turns the identified coordinate into a closed-form table. Its inverse-CDF quantiles follow from a stated convex surrogate, require no learned positional parameters, and are unique for that surrogate.

We test the same coordinate in three complementary regimes: fully frozen checkpoints, pretrained models under matched low-rank adaptation, and from-initialisation or continued full-parameter training. They ask whether z acts without weight updates, can be exploited after limited adaptation, and is learned through weight-table co-adaptation, respectively. Full and scarce rotary budgets and a bidirectional 3D video DiT extend these tests across architectures and beyond autoregressive text (§4).

Contributions. (i) We expose z as a third finite-RoPE coordinate—phase-resolution placement within fixed support and K —and causally identify its effect in a paired three-seed training intervention; an independent frozen-checkpoint pure- z intervention shows that the same coordinate remains active after pretraining. (ii) We give an exact phase-invariant account of the full sin/cos basis: a spectral-budget identity, a slow-frequency collapse law, the weight-table co-adaptation boundary, and a frozen-transplant obstruction. (iii) We derive EVQ-COSH as a closed-form, zero-learned-parameter construction for a stated convex surrogate and demonstrate control through fully frozen, matched-adaptation, and from-training interventions.

2 IDENTIFYING THE ALLOCATION AXIS

2.1 FIXED-SUPPORT TRAINING INTERVENTION

Allocation is identifiable at fixed support. Within each of three 151.9M training seeds, both arms share architecture, initialisation, token order, optimiser, LR schedule, global batch, a 499,974,144-token budget, and 32 evaluation anchors. The baseline is FMRoPE at fixed training support, implementing Oka et al. (2026b, §6.1) with $\theta=L_{\text{train}}=256$ and geometric exponents. Anchored EVQ-COSH ($\tau=4$) normalises its quantiles to the same extrema and log-span, so only the $K-2=30$ interior frequencies move while retaining the deployed Cosh table’s z . Every training seed favours anchored EVQ-COSH at every OOD length. The anchored EVQ-COSH minus FMRoPE NLL difference is +0.026 at the 256-token training length and $-0.281/-0.176/-0.146$ at 512/1K/2K (Fig. 1c). No geometric base can reproduce this intervention once the sampled support is pinned:

fixed support + different $z \Rightarrow$ different trained-model behaviour. Thus z is a separately identifiable coordinate, not a disguised base change. Retargeting the support reverses the tested ordering, showing that support and allocation are distinct but interacting design coordinates; the full protocol and per-seed contrasts are in App. C.

Configuration and shape breadth. A pre-specified 50.9M factorial versus Geo spans two bases, two training lengths, three head dimensions, three paired seeds, scaled Cosh schedules, and a deformation-matched exponential (App. Table 9). It supplies configuration and shape breadth across twelve structural settings; model-scale persistence is tested separately in §4.3.

3 THEORY

The interventions establish that interior allocation changes model behaviour; theory explains how such a within-support change can matter. Exact geometry of the full sin/cos pairs shows how z redistributes a fixed spectral budget and why geometric slow bands can collapse onto nearly the same positional basis. Weight-table co-adaptation and the transplant obstruction explain why the installation stage matters. A stated convex surrogate then yields EVQ-COSH as an analytic construction on the same coordinate.

3.1 A FINITE TABLE IS A SET OF TWO-DIMENSIONAL SUBSPACES

Let $\phi \in [0, 1]$ parameterise log-frequency, $\omega(\phi) = b^{-\phi}$, let d_{rot} be the number of rotated dimensions and $K = d_{\text{rot}}/2$ the number of rotary pairs ($d_{\text{rot}} = d_{\text{head}}$ for standard MHA, smaller for compressed-RoPE architectures such as MLA).

We reserve z for the support-normalised coordinate in Eq. (1). Standard implementations use raw geometric exponents $u_k = k/K$, while deployed EVQ-COSH uses midpoint quantiles $u_k = (k + 1/2)/K$; endpoint-inclusive grids used for static diagnostics are labelled explicitly. Normalising the sampled geometric endpoints gives $z_k = k/(K - 1)$.

Equation (1) is a coordinate identity for any ordered positive table with nonzero span. In the geometric family $z_k = k/(K - 1)$, so fixing (a, R) fixes every channel. A second table with the same support and different interior z therefore cannot be represented by a different scalar base. Quantities such as median wavelength or the number of channels completing a cycle are summaries of z , not omitted scalar controls.

The frozen-checkpoint experiments instantiate this identity without using the Cosh family. Given Native frequencies ω_k , an extension factor s , and channel-wise displacement weights $m_k \in [0, 1]$, they form

$$\hat{\omega}_k = \omega_k[(1 - m_k) + m_k/s], \quad \hat{z}_k = \frac{-\log \hat{\omega}_k - \hat{a}}{\hat{R}}, \quad (2)$$

with the fast and slow endpoints pinned, so $\hat{a} = -\log \hat{\omega}_0$ and $\hat{R} = -\log \hat{\omega}_{K-1} - \hat{a}$. The uniform-allocation control sets $\hat{z}_k = k/(K - 1)$. For the derived arm, $\hat{U}_k$ is the normalised residual-energy fraction left after the full sin/cos pair is projected onto all other pairs over Native separations weighted by their causal pair count, $p(\Delta) \propto L - \Delta$. It sets $m_k = (1 - \hat{U}_k)^2$, so less unique pairs move farther. A coarse, label-free control fits a linear ramp to these displacements without using task labels. All three tables have the same K and endpoints; after normalisation they are simply three values of the same interior coordinate z .

A rotary pair contributes $f_\omega(\Delta) = C \cos(\omega\Delta) + D \sin(\omega\Delta)$ to the relative-position attention logit, with C, D set by content. The object attached to ω is therefore the two-dimensional subspace

$$V_\omega = \text{span}\{\cos(\omega\Delta), \sin(\omega\Delta)\}, \quad (3)$$

not a cosine feature: an analysis that fixes one content phase sees a slice of V_ω and is not invariant to rotation inside the pair. Two channels are redundant to the extent that their subspaces are. With $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, Gram matrices $S_\omega = \mathbb{E}[x_\omega^\top x_\omega]$ and $H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu]$ over a separation prior, and the whitened cross-Gram $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ whenever $S_\omega, S_\nu \succ 0$, the singular values of $Q_{\omega\nu}$ are the canonical correlations between V_ω and V_ν , and

$$c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2 \in [0, 1] \quad (4)$$

is invariant to phase rotation inside a pair and to any change of basis of either subspace. For $\Delta \sim \text{Unif}[0, L]$, $H_{\omega\nu}$ has a closed form in $\sin t/t$ and $(1 - \cos t)/t$ at $t = (\omega \mp \nu)L$ (App. A).

3.2 THE SPECTRAL BUDGET IS AN EFFECTIVE-DIMENSION ACCOUNT

Block-whitening every pair gives a global correlation Gram Γ with diagonal blocks I_2 . Write $\bar{c}$ for the mean of the c_{ij} .

Theorem 1 (Spectral budget identity). $\text{tr } \Gamma = 2K$ and $\text{tr}(\Gamma^2) = 2K[1 + (K-1)\bar{c}]$, so the Rényi-2 effective rank of the positional basis is exactly

$$r_2(\Gamma) = \frac{(\text{tr } \Gamma)^2}{\text{tr}(\Gamma^2)} = \frac{2K}{1 + (K-1)\bar{c}}. \quad (5)$$

The denominator turns the spectral budget into an exact Rényi-2 effective-dimension identity: larger mean pairwise redundancy lowers r_2 . Because $\text{tr } \Gamma = 2K$ is fixed, allocation cannot create nominal rotary dimension; it changes how evenly that fixed budget is distributed across positional directions. This is the precise sense in which a finite RoPE table has a spectral budget. Fix K , sampled support (a, R) , and separation measure μ . Substituting $\omega_k(z) = \exp[-(a + Rz_k)]$ into the full-pair block-whitened positional Gram $\Gamma_\mu(z)$ makes its spectrum, and hence the effective sin/cos basis realised by the nominal rotary budget, a function of interior allocation. Geometry diagnoses that basis; the fixed-support intervention of §2.1 establishes its behavioural consequence. Block whitening removes per-channel energy and conditioning from the diagnosis.

The deflation is concentrated at the slow end.

Proposition 2 (Low-frequency collapse). As $\omega L \rightarrow 0$, $V_\omega \rightarrow \text{span}\{1, \Delta\}$. For slow channels $x=\omega L$, $y=\nu L$, and $\epsilon = \max\{|x|, |y|\} \rightarrow 0$, $2 - \|Q_{x,y}\|_F^2 = \frac{19}{12600}(x^2 - y^2)^2 + O(\epsilon^6)$: slow subspaces approach a common two-dimensional limit at fourth order. Under the softmax metric for a fixed p with nondegenerate support on at least three distances, $F = \text{diag}(p) - pp^\top$ annihilates the constant direction, and the centred limit is $\text{span}\{\Delta - \mathbb{E}_p \Delta, \Delta^2 - \mathbb{E}_p \Delta^2\}$.

Concretely, at $L=4096$, $b=5 \times 10^5$, $K=64$ the 23 pairs with $\omega L \leq 1$ occupy 46 nominal dimensions but have block-whitened $r_2 = 2.00$. Thus more than one third of the rotary pairs spend 46 nominal dimensions on approximately two effective positional dimensions. The channels may still carry content; the geometric redundancy tax is that many of them describe almost the same positional directions. Proposition 2 characterises the severe slow-band limit. The fixed-support identification also covers settings outside that asymptotic regime: in its $b=256$ control no sampled pair lies deep in the limit, and Theorem 1 measures overlap over the full table. Additional allocation and redundancy views appear in App. Fig. 2.

3.3 THE TABLE IS A TRAINING-TIME COORDINATE SYSTEM

Theorem 1 measures the geometry available to a table; trained behaviour additionally depends on the coefficients learned in that basis. To separate the two, we cross two trained 50M weight sets with two runtime tables at $L=512$, $b=5 \times 10^5$, and $\tau_{\text{EVQ}}=2.83$. All parameters remain frozen throughout this diagnostic (Table 1). Substituting EVQ-COSH for the geometric table at inference raises the block-whitened r_2 from 4.57 to 12.54 while perplexity degrades from 7.14 to 76.20.

Two consequences discipline the rest of the paper. First, collision, effective rank, and log-determinant describe the table without its learned coefficients; the task claims in §4 are therefore established on trained models. Second, the two *self-consistent* systems are essentially tied in-window (7.14 vs. 7.16) while both mismatched cells collapse: on LM loss the table and weight main effects are $+0.599$ and -0.597 , and their interaction contrast is -3.537 (configuration-level bootstrap CI $[-5.17, -3.04]$). The table behaves as a coordinate system fixed before training, with the weights learning coefficients in it.

Theorem 3 (Post-hoc transplant obstruction). Let $\mathcal{R}_\Omega(\Delta)$ be the block-rotation operator of a frequency multiset Ω . If position-independent invertible maps A, B satisfy $A^\top \mathcal{R}_{\Omega'}(\Delta)B = \mathcal{R}_\Omega(\Delta)$ for all Δ in an open interval containing zero, then Ω' and Ω have the same frequency multiset, up to sign and permutation. Repeated frequencies may mix within their full equal-frequency invariant subspace.

Table 1: **Weights co-adapt to their training-time frequency table.** Two seed-42 50M models trained with the geometric and EVQ-COSH tables, each evaluated under both runtime tables with all parameters frozen (1,920 head–query observations). Both runtime tables use the shared midpoint grid $u_k = (k + 1/2)/K$ for this diagnostic. The matched pairs remain near PPL 7, whereas both cross-swaps fail; the worst cell is also the one whose static r_2 improves most.

Trained weights	Runtime table	PPL	Rényi-2 rank r_2
Geo	Geo	7.14	4.57
Geo	EVQ-COSH	76.20	12.54
EVQ-COSH	Geo	23.05	4.57
EVQ-COSH	EVQ-COSH	7.16	12.54

The proof is short: $\Delta=0$ gives $A^\top B = I$, so the relation becomes a similarity $A^\top \mathcal{R}_{\Omega'} A^{-\top} = \mathcal{R}_{\Omega}$; differentiating at zero makes the generators similar, and similarity preserves the spectrum $\{\pm i\omega_k\}$ (App. A). The theorem therefore characterises exact function preservation by this broad class of fixed linear Q/K compensation. The finite-distribution protocols of §4.2 measure the complementary case in which Q/K or broader low-rank paths adapt to the new table.

3.4 AN ANALYTIC NON-UNIFORM ALLOCATION FAMILY

A positive density $\rho \in C^2([0, 1])$ with $\int_0^1 \rho = 1$ induces a discrete allocation by inverse-CDF quantisation, $\phi_k = F_\rho^{-1}(u_k)$; geometric RoPE is $\rho \equiv 1$. The exact canonical-correlation objective is oscillatory. To obtain a closed-form construction, we use the convex modelling surrogate

$$\mathcal{C}_{\text{app}}[\rho] = \frac{\alpha}{2} \int_0^1 \rho^2 d\phi + \frac{\beta}{2} \iint_{[0,1]^2} \rho(\phi)\rho(\psi) \min(\phi, \psi) d\phi d\psi, \quad (6)$$

with $\alpha > 0, \beta \geq 0$. The diagonal term penalises channel-load concentration; $\min(\phi, \psi)$ is the Green kernel of $-\partial_\phi^2$ and encodes a penalty for pairs of channels that are both slow, matching the direction of Prop. 2. The surrogate supplies the analytic allocation; phase-invariant geometry measures the resulting basis. Under the closed-form cosine-feature kernel, the derived table lowers collision by 24–92% and raises its cosine-Gram Shannon rank in all 12 tested configurations (App. A.5), while trained interventions establish the behavioural effect.

Theorem 4 (Closed-form optimum under the surrogate). *Under (6) with $\alpha > 0, \beta \geq 0$ and constraints $\rho \in C^2([0, 1])$, $\rho > 0$, $\int_0^1 \rho = 1$, the unique constrained minimiser is*

$$\rho_\tau(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad \tau = \sqrt{\beta/\alpha}; \quad (7)$$

at $\beta=0$ it is the $\tau \rightarrow 0$ limit $\rho_0 \equiv 1$.

Corollary 5 (Closed-form quantiles). $F_\tau(\phi) = 1 - \sinh(\tau(1-\phi))/\sinh \tau$, so

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau), \quad u_k = \frac{k+1/2}{K}, \quad (8)$$

which tends to u_k as $\tau \rightarrow 0$, recovering the uniform-log midpoint grid.

The theorem selects the Cosh family exactly for (6); midpoint quantisation turns that density into a finite RoPE table. For $\tau > 0$, ρ_τ decreases across log-frequency: relative to the uniform grid it places more samples on the fast side and thins the redundant slow side, while $\tau \rightarrow 0$ recovers the geometric allocation. The inverse CDF therefore discretises the continuous solution without search or learned positional parameters. Its uniqueness is conditional on the stated surrogate. The pre-specified operating rule, finite multiplier evidence, and grid conventions are in App. A.8. The exact-range endpoint embedding is specified in App. C.

Viewed together, the theory turns finite RoPE design into a basis-allocation problem. The coordinate z selects the basis available to training; the collapse law identifies where geometric sampling spends the finite budget redundantly; the crossing and obstruction show that learned coefficients bind to the installed basis; and the variational solution supplies a table before those coefficients are learned. Section 4 tests the same coordinate with weights fully frozen, under matched low-rank adaptation, and through from-training co-adaptation.

4 THE ALLOCATION COORDINATE ACROSS MODEL-BUILDING REGIMES

Three intervention regimes test how deeply weights must change before a model can use a new z . The frozen route asks whether allocation alone changes a mature model; matched low-rank adaptation asks whether pretrained representations can exploit a new basis with limited updates; from-training studies ask whether weights learn and use that basis through co-adaptation. We present them in that order, beginning with the direct mature-model intervention.

4.1 FULLY FROZEN: ZERO-TRAINING INTERVENTION

Allocation alone changes a mature model. Using $s=4$, Eq. (2) constructs alternative tables with the same fast and slow endpoints; all arms also share the checkpoint, attention amplitude, evaluation rows, decoder, precision, and hardware. The OLMo table, amplitude, split rule, checkpoint, and metric were frozen before the nine confirmation tasks were evaluated. Table 2 separates standard references from the matched-support block. On OLMo, derived minus uniform allocation is +59.92 points ([54.88, 64.80]), and derived minus official Transformers YaRN is +52.54 points ([46.99, 57.95]). The common amplitude $1 + 0.1 \ln 4$ is YaRN’s default factor-four attention scaling. On Qwen at 64K, derived reaches 66.50% versus 57.75% for uniform allocation and 60.25% for YaRN. The model-relative derivation locates where the allocation should bend; a coarse label-free control fitted to it reproduces the OLMo recovery at 61.04% (App. E.2).

Table 2: **Frozen mature-checkpoint RULER macro (%)**. OLMo is evaluated at $4\times$ its Native window and Qwen at $2\times$. Reference methods are separated from the fixed-support block. Only the lower block holds support, amplitude, checkpoint, rows, decoder, precision, and hardware fixed while changing interior z . Each task has 20 rows; paired intervals condition on each fixed checkpoint and task set through evaluation-row resampling.

Installed table	OLMo unseen-nine, 16K	Qwen core-four, 64K
Native	0.00	54.50
Official Transformers YaRN ($s=4$)	7.94	60.25
Uniform allocation z	0.56	57.75
Coarse label-free z	61.04	64.00
Derived allocation z	60.47	66.50

Zero-training deployment. Building on the pure- z effect, the zero-training method installs the reallocated long table with a fixed amplitude; a Native/long session route is also evaluated as a deployment option. The system adds no learned parameters or training tokens.

Natural-text likelihood. On FineWeb-Edu, the long profile at 4K gives NLL 2.8774 versus 2.7538 for Native, a difference of +0.1236 NLL. Against official Transformers YaRN at factor four, PG-19 tail NLL improves from 3.4404/3.7955 to 3.1060/3.0977 at 8/16K.

Task capability. On 13-family RULER at 8/16K, macro is 0/0% for Native, 23.82/5.88% for official Transformers YaRN at factor four, and 67.72/54.40% for the zero-training policy, which is higher on 12 of 13 families at each length. On full-200 Qasper, the session policy raises token F1 from 18.03% to 24.57%; on full-200 2Wiki, applying the long profile to every row raises token F1 from 25.69% to 27.74% (the routed policy scores 26.66%). Full protocols are in App. E.2.

4.2 PRETRAINED REPRESENTATIONS: MATCHED LOW-RANK ADAPTATION

Limited adaptation asks whether a mature pretrained model can learn to use a new allocation without full-parameter retraining. The 1.485B studies measure task-family length transfer; the 8B source-blocking intervention tests whether the adapted model causally uses remote evidence.

1.485B task-family length transfer. Starting from protocol-matched Native and EVQ-COSH Stage-A parents, both OLMo-2 arms freeze inherited V/O LoRA and continue only Q/K for 300

steps with identical data, order, optimiser, trainable count, and phase exposure. On 200 held-out 2WikiMultiHopQA rows per length, Native/EVQ-COSH token-F1 is 25.99/24.84% at 4K, 0.07/21.48% at 8K, and 0/8.57% at 16K (Ho et al., 2020; Bai et al., 2024). An independent 13-family RULER continuation gives Native/EVQ-COSH macro 72.19/42.44% at 4K, 2.02/31.63% at 8K, and 0.38/5.03% at 16K (Hsieh et al., 2024). Training and evaluation use disjoint rows from the same task families, establishing task-family-adapted length transfer (App. E).

8B causal remote-source use. On LLaMA-3-8B-Instruct (Grattafiori et al., 2024), matched rank-64 Q/K/V/O LoRA gives Native/EVQ-COSH PPL 6.82/10.07, 108.96/24.07, and 991.48/127.91 at 8/16/32K, with the long-range direction holding across all 24 frozen natural-text packs and three domains. At true 16K, median target-block hit@16 rises from 18.75% to 64.06%; blocking decode attention to the remote gold block changes NLL by $-0.0095/ + 1.5055$. This establishes causal remote-source use in the tested adapted model (App. E).

4.3 FROM TRAINING: WEIGHT-TABLE CO-ADAPTATION

Installing z before or during full-parameter training asks whether weights learn to exploit the resulting positional basis. The paired three-seed result in §2.1 supplies the fixed-support causal anchor; the experiments below test its persistence across architecture, scale, and modality.

Scarce-channel architecture. Interior allocation remains consequential when the rotary budget is only $K=16$: a 432M MLA-style transformer (DeepSeek-AI, 2024) trained from scratch over three seeds gives Geo/EVQ-COSH PPL 35.4/35.8 at the 8K training length and 138.8/95.6 at 16K, with every seed favouring EVQ-COSH (App. Table 12).

Full-parameter persistence and early-training scale. Starting both 750M arms from a shared 2K Geo checkpoint and continuing all parameters for 500M tokens at 4K gives Geo/EVQ-COSH PPL 22.0/22.3 at 4K but 45.1/24.4 at 16K; strict autoregressive exact retrieval at 8K changes from 0% to 77.5% (App. Table 6). At 1.485B, a pre-specified early-training comparison starts from the same public initialisation, follows the same scientific recipe, and uses a matched 2.097B-token budget. EVQ-COSH crosses by 8K, with 122/128 and 126/128 held-out documents favouring it at 8/16K (App. Fig. 5). Together these protocols repeat the in-window/long-length crossover through 1.485B.

Modal breadth. On Oscillating Moving MNIST, a seed-42 head-to-head comparison in a 129.6M video DiT with bidirectional 3D RoPE gives Geo-to-EVQ-COSH denoising-MSE deltas of -21% on train, -16% over all extrapolated frames, and -35% on far extrapolated frames (App. Table 7), extending the evidence beyond autoregressive text.

5 RELATED WORK

Positional encoding can determine whether a transformer generalises beyond its training length (Kazemnejad et al., 2023). RoPE extensions rescale positions, change a scalar base, transport frequency bands, or remap remote phases (Chen et al., 2023; Peng et al., 2024; Ding et al., 2024; Shang et al., 2025; Oka et al., 2026b; Tian et al., 2026; Tang et al., 2026). Frequency-dependent transport can realise a non-uniform z , but couples it to support or phase changes. Base analyses bound the geometric family’s attainable context and identify a critical dimension (Xu et al., 2024; Liu et al., 2024); learned use of a supplied grid adds a data-dependent view (Wu et al., 2026). We isolate z directly: what changes when both endpoints and the log-span are pinned and only its interiors move?

Other work measures unequal use of RoPE bands, removes or clips selected rotations, learns input-dependent angles or frequency tables, or replaces the fixed table with a larger positional function (Barbero et al., 2025; Oka et al., 2026a; Chiang & Yogatama, 2025; Chen et al., 2025; Li et al., 2026a; Wertheimer et al., 2026; Zhang et al., 2026; Movahedi et al., 2026; Wang et al., 2024; Zheng et al., 2024; Li et al., 2024; Hua et al., 2025; Li et al., 2026b). LeRoPE and AdaRoPE make allocation explicit through learned non-geometric frequencies (Karypis et al., 2026; Wang et al., 2026); LeRoPE’s frozen-table ablation also makes the in-window value of allocation visible. We retain all K canonical rotary planes, identify z at fixed support, and derive a closed-form table without frequency search or learned positional parameters.

6 DISCUSSION

A finite RoPE table is specified by its sampled log-frequency support (a, R) and the interior allocation z of its fixed K pairs. Position rescaling or frequency-dependent transport may change the realised table as context grows, but does not remove this decomposition. Our fixed-support intervention holds K , both endpoints, and the RoPE operator constant and changes only z —a design move outside a scalar-base sweep.

The three empirical routes intervene on this coordinate at different stages. With weights fully frozen, z alone changes mature-model behaviour; under matched low-rank adaptation, pretrained representations learn to use a new basis; from training, the paired fixed-support control identifies weight-table co-adaptation. Exact full-subspace geometry explains how those tables differ as positional bases, while the weights-by-table crossing and transplant obstruction explain why the intervention stage matters.

Across the autoregressive text protocols that evaluate both the original training or Native window and longer positions, the comparisons reveal a within-protocol crossover: the baseline is stronger near the original window, while the reallocated table overtakes it beyond that window. This recurring behavioural signature of finite-budget reallocation directly motivates the Native/long session route. The video DiT instead improves both training-frame and extrapolated-frame MSE; its pre-specified diagnostic locates the different regime: six of sixteen temporal channels accumulate less than 0.1 rad over the 32 training frames, so reallocating them changes in-window phase coverage rather than imposing the text-model crossover.

Extrapolation provides the sharpest present identification of z . The same coordinate also changes which phase scales are available inside the original window. LeRoPE reports in-window gains from learned non-geometric tables; our fixed-support geometry and closed-form construction provide complementary ways to analyse and instantiate the same broader design space. Together they make joint native-window and long-range allocation a natural next design objective.

The practical consequence is that a finite RoPE table should be specified by both its sampled support and its interior allocation. A shared base, endpoint, or pair count does not by itself define a matched finite table. Treating z as an explicit design object makes allocation measurable and controllable across training, adaptation, and deployment; the exact geometry supplies the basis diagnosis, and the controlled interventions establish its behavioural consequences.

ETHICS STATEMENT

This work introduces no new dataset, deployed system, or human-subject component. The proposed training-time construction changes only the RoPE inverse-frequency table and adds no learned parameters; the mature-checkpoint studies use the frozen-table or LoRA interventions stated in their protocols. The table construction remains inexpensive to audit, ablate, or revert. All models, corpora, and evaluation suites used here are public research artefacts used within their stated licences.

A better allocated frequency table may reduce the adaptation or inference effort required to reach a target context range, and the same mechanism could be used to extend the context capability of language or video models, with the general risks associated with stronger generative models. Our evaluations measure positional and long-context endpoints, not factual reliability or safety; deployment claims require application-specific evaluation.

REPRODUCIBILITY STATEMENT

Equation (8) specifies the default midpoint EVQ-COSH table as a pure function of (τ, K) , with no learned parameters or auxiliary loss. Anchored support embeddings, endpoint $u_k = k/K$ quantisation in the mature OLMo EVQ-COSH protocols, and the inclusive static diagnostic are labelled where used; each retains the same continuous Cosh allocation law but defines a different finite grid or support. Equation (22) is the full-RoPE EVQ-COSH reference rule; every EVQ-COSH protocol reports its realised τ explicitly, including the separately fixed MLA and video settings. The mature zero-training uniform/coarse/derived allocation tables instead follow Eq. (2) and are specified in App. E. Complete proofs of Theorems 1, 3 and 4, Proposition 2, Corollary 5 and Lemma 7, together with every stated assumption, are in App. A. The identification protocol, including how sampled

extrema and log-span are pinned, the exact pre-specified factorial design, and the corpora used by each arm, is in App. C; the mature-scale adaptation protocols, module budgets, phase-exposure mixtures, and evaluation harness settings are in App. E; remaining architectures and hyperparameters are in App. B. Seed scope is stated with every result. The anonymous supplement includes the exact-range training entrypoint and three-seed aggregate, the frozen figure generators, curated mature-protocol snapshots, and CPU tests, with a reviewer-facing file map and commands.

AI USE STATEMENT

We used generative AI tools to help develop conceptual frameworks and mathematical claims; propose and refine hypotheses; draft and check proofs; give feedback on methodology and experiments; implement methods and deterministic data-reformatting and analysis code; translate or rephrase research notes; and interpret results. We did not use generative AI to generate dataset content or to perform qualitative or thematic data analysis; survey, interview, and transcription tasks do not apply to this work.

For tasks for which disclosure is recommended, we also used these tools to suggest experimental parameters; write, edit, and test software and artefacts; create or modify figures; draft, edit, and structure parts of the manuscript; brainstorm and identify research gaps; search for and summarise literature; and format references. Every empirical quantity reported here comes from executed code over recorded run artefacts rather than language-model generation.

All AI-assisted work was reviewed before inclusion. We take responsibility for the final text, claims, proofs, code, figures, and experimental artefacts.

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A PROOF DETAILS

A.1 SUBSPACE GEOMETRY: CROSS-GRAM, BUDGET IDENTITY, AND COLLAPSE

Closed-form cross-Gram. For $\Delta \sim \text{Unif}[0, L]$ and $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, write $d = (\omega - \nu)L$, $s = (\omega + \nu)L$, $a(t) = \sin t/t$ and $b_*(t) = (1 - \cos t)/t$. Expanding the products of sinusoids and integrating termwise gives

$$H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu] = \frac{1}{2} \begin{bmatrix} a(d) + a(s) & b_*(s) - b_*(d) \\ b_*(s) + b_*(d) & a(d) - a(s) \end{bmatrix}, \quad (9)$$

and $S_\omega = H_{\omega\omega}$. Since $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ is the whitened cross-Gram of two two-dimensional subspaces, its singular values are the canonical correlations, and $c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2$ is invariant under $x_\omega \mapsto x_\omega O$ for any invertible O (in particular under rotation of the phase inside a pair), because S_ω transforms congruently and the whitening cancels O .

Basis invariance. The cancellation can be stated exactly. Under independent invertible basis changes $x_\omega \mapsto x_\omega A_\omega$ and $x_\nu \mapsto x_\nu A_\nu$, let $U_j = S_j^{1/2} A_j (A_j^\top S_j A_j)^{-1/2}$ for $j \in \{\omega, \nu\}$. Then $U_j^\top U_j = I$, and the transformed whitened cross-Gram is $\tilde{Q}_{\omega\nu} = U_\omega^\top Q_{\omega\nu} U_\nu$. Left and right orthogonal factors preserve the two canonical correlations and therefore $c_{\omega\nu}$. The metric is consequently attached to the positional subspaces themselves, not to a chosen sine/cosine phase convention.

Proof of Theorem 1. Γ is the block matrix with blocks $\Gamma_{ij} = Q_{\omega_i \omega_j}$ and $\Gamma_{ii} = I_2$. Hence $\text{tr} \Gamma = \sum_{i=1}^K \text{tr} I_2 = 2K$. For the second moment, $\text{tr}(\Gamma^2) = \sum_{i,j} \text{tr}(\Gamma_{ij} \Gamma_{ji}) = \sum_i \text{tr}(I_2) + \sum_{i \neq j} \|Q_{\omega_i \omega_j}\|_F^2 = 2K + 2K(K-1)\bar{c}$, using $\Gamma_{ji} = \Gamma_{ij}^\top$ and the definition $\bar{c} = \frac{1}{K(K-1)} \sum_{i \neq j} c_{\omega_i \omega_j}$ with $\|Q_{ij}\|_F^2 = 2c_{ij}$. Dividing gives (5). $\square$

The identity determines Rényi-2 rank exactly from the mean pairwise redundancy. Shannon effective rank and log-determinant retain higher-order multi-subspace dependence on the full spectrum.

Proof of Proposition 2. With $t = \Delta/L$ and $x = \omega L$, $\cos(xt) = 1 - x^2 t^2/2 + O(x^4)$ and $\sin(xt)/x = t - x^2 t^3/6 + O(x^4)$, so after rescaling the second coordinate the basis of V_ω tends to $\{1, \Delta\}$. Substituting these expansions into (9) and whitening, the leading deficit between two slow subspaces is $2 - \|Q_{x,y}\|_F^2 = \frac{19}{12600}(x^2 - y^2)^2 + O(\epsilon^6)$, where $\epsilon = \max\{|x|, |y|\} \rightarrow 0$; numerically at $x=0.05, y=0.10$ the exact-to-leading ratio is 1.00058. For the softmax metric with fixed p , $F = \text{diag}(p) - pp^\top$ satisfies $F\mathbf{1} = 0$, so constants are annihilated; centring by $\mathbb{E}_p$ and rescaling gives $\sin(\omega\Delta)/\omega \rightarrow \Delta - \mathbb{E}_p \Delta$ and $-2\cos(\omega\Delta)/\omega^2 \rightarrow \Delta^2 - \mathbb{E}_p \Delta^2$, which span the centred limit whenever p has nondegenerate support on at least three distances. $\square$

The reported $r_2=2.00$ for the 23 slow pairs uses the standard $u_k=k/K$ grid and the uniform separation prior above. Other priors change that static numerical Gram. The fixed-range trained-model contrast of §2.1 instead follows its paired empirical protocol.

A.2 EFFECTIVE DIMENSION AND MULTISCALE COVERAGE

The budget identity is an exact account of positional dimension under its separation prior. A second requirement is where that dimension is placed across scales. The distinction is already visible in an exact construction.

Lemma 6 (Full in-window rank can coexist with exact out-of-window aliasing). *Let $n_1, \dots, n_K$ be distinct positive integers and set $\omega_k = 2\pi n_k/L$. Under $\Delta \sim \text{Unif}[0, L]$, the block-whitened correlation Gram of the corresponding full sin/cos pairs is $\Gamma = I_{2K}$, hence $r_2(\Gamma) = 2K$. Nevertheless the complete positional feature map satisfies $\Phi(\Delta + L) = \Phi(\Delta)$ for every Δ .*

Proof. Fourier orthogonality on $[0, L]$ makes the sine and cosine functions at distinct integer harmonics mutually orthogonal, while each nonzero harmonic has self-Gram $\frac{1}{2}I_2$. Block whitening therefore produces the identity Gram and attains the full nominal rank. For every k , $e^{i\omega_k(\Delta+L)} = e^{i\omega_k\Delta} e^{i2\pi n_k} = e^{i\omega_k\Delta}$, proving the period- L alias for all pairs jointly. $\square$

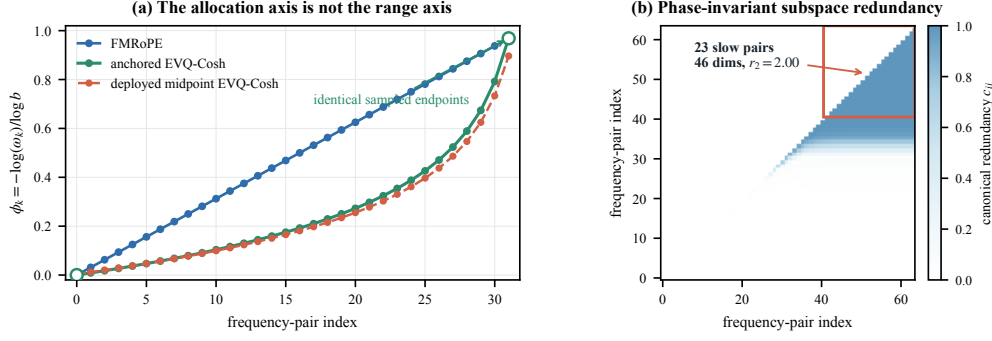


Figure 2: **Where a finite frequency table spends its budget.** (a) At $K=32$, Cosh concentrates interior channels toward the fast end while endpoint normalisation preserves sampled support; the deployed midpoint table is shown separately. (b) For standard RoPE at $L=4096$, $b=5 \times 10^5$, and $K=64$, phase-invariant full-subspace redundancy concentrates in the 23 pairs with $\omega L \leq 1$.

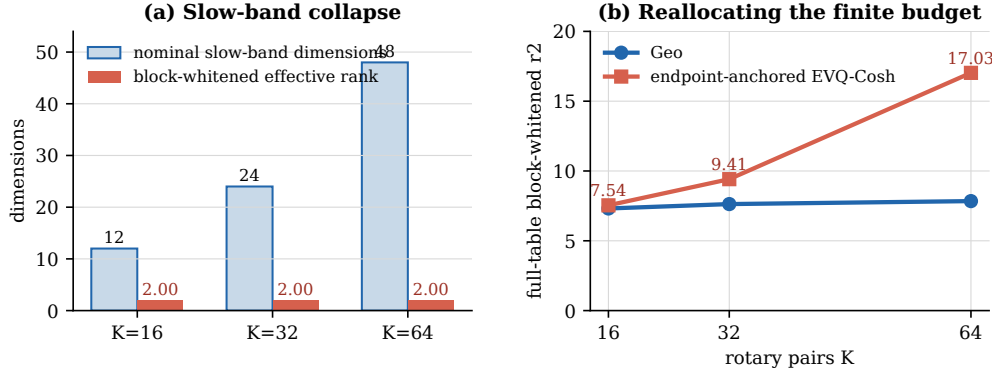


Figure 3: **Spectral-budget anatomy across channel counts.** Panel (a) uses the endpoint-inclusive diagnostic grid at $L=4096$ and $b=500K$: as K grows from 16 to 64, the nominal dimension assigned to pairs with $\omega L \leq 1$ grows from 12 to 48, while their phase-invariant effective rank remains approximately two. Panel (b) evaluates the complete fixed-endpoint table under the same uniform separation prior, using the zero-search reference $\tau=2K/\sqrt{L}$; anchored EVQ-COSH spends progressively more of the available full-pair dimension than Geo as the budget grows. These are static positional-basis diagnostics; the trained effect is identified separately in App. C.

Lemma 6 separates two design questions. Theorem 1 measures how many distinct positional directions a finite table supplies over the chosen interval; multiscale coverage is a separate criterion for how those directions span separations beyond one interval. The fixed-support experiments evaluate that allocation with trained models, while Table 1 shows why the learned coefficients must be kept with the basis that trained them.

A.3 POST-HOC TRANSPLANT OBSTRUCTION

Proof of Theorem 3. Both $\mathcal{R}_\Omega$ and $\mathcal{R}_{\Omega'}$ are block-diagonal with 2×2 rotation blocks, hence $\mathcal{R}_\Omega(0) = \mathcal{R}_{\Omega'}(0) = I$. Evaluating the hypothesis at $\Delta=0$ gives $A^\top B = I$, so $B = A^{-\top}$ and the hypothesis becomes the similarity $A^\top \mathcal{R}_{\Omega'}(\Delta) A^{-\top} = \mathcal{R}_\Omega(\Delta)$ on an interval. Each side is the exponential of a constant generator, $\mathcal{R}_\Omega(\Delta) = \exp(\Delta G_\Omega)$ with $G_\Omega = \bigoplus_k \omega_k J$, $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Differentiating at $\Delta=0$ gives $A^\top G_{\Omega'} A^{-\top} = G_\Omega$, so the generators are similar and share a spectrum. The spectrum of G_Ω is $\{\pm i\omega_k\}_{k \leq K}$ with multiplicity, so the frequency multisets agree up to sign and permutation. When a frequency is repeated, similarity may mix the entire equal-frequency invariant subspace; it need not preserve the original two-dimensional block decomposition. For integer positions the same argument is replaced by a one-step spectral argument: evaluating at $\Delta = 0$ again gives $B = A^{-\top}$,

while $\Delta = 1$ makes $\mathcal{R}_{\Omega'}(1)$ and $\mathcal{R}_{\Omega}(1)$ similar. Their eigenvalue multisets $\{e^{\pm i\omega_k}\}$ therefore agree, which identifies frequencies up to sign, permutation, and the 2π alias. $\square$

This exact statement covers *fixed, position-independent, invertible* compensation. Approximate retraining on finite data, nonlinear maps, and changes to V/O or the residual path are separate intervention classes.

A.4 PROOF OF THE CLOSED-FORM VARIATIONAL OPTIMUM

This subsection proves Theorem 4 directly from the stated variational objective. Starting from the kernel

$$K_{\text{app}}(\phi, \psi) = \alpha\delta(\phi - \psi) + \beta \min(\phi, \psi), \quad (10)$$

the main-text collision functional is (6), which under the mass normalization $\int_0^1 \rho = 1$ with Lagrange multiplier ν has constrained first variation

$$\alpha\rho(\phi) + \beta g(\phi) + \nu = 0, \quad g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi. \quad (11)$$

Existence. For rigor, view $\mathcal{C}_{\text{app}}$ first as a functional on $L^2([0, 1])$ and minimize over $\mathcal{A} = \{\rho \in L^2([0, 1]) : \rho \geq 0 \text{ a.e., } \int_0^1 \rho = 1\}$. The $\alpha\|\rho\|_2^2$ term is coercive and weakly lower semicontinuous, the Green-kernel term is positive semidefinite and continuous, and the mass/nonnegativity constraints are weakly closed. A minimizer therefore exists by the direct method. Strict convexity gives uniqueness. The Euler solution derived below is strictly positive, so the nonnegativity constraint is inactive a posteriori and the classical C^2 derivation is justified.

Direct differentiation of g gives

$$g'(\phi) = \int_{\phi}^1 \rho(\psi) d\psi, \quad g''(\phi) = -\rho(\phi). \quad (12)$$

Differentiating the first variation twice in ϕ eliminates ν (a constant) and yields the homogeneous ODE

$$\rho''(\phi) - \tau^2 \rho(\phi) = 0, \quad \tau = \sqrt{\beta/\alpha}. \quad (13)$$

Evaluating the pre-differentiated stationarity together with $g(0)=0$, $g'(1)=0$, and $\int_0^1 \rho = 1$ gives the derivative boundary conditions

$$\rho'(0) = -\tau^2, \quad \rho'(1) = 0, \quad (14)$$

since differentiating $\alpha\rho(\phi) + \beta g(\phi) + \nu = 0$ once and evaluating at $\phi=1$ gives $\alpha\rho'(1) + \beta g'(1) = 0$ hence $\rho'(1)=0$, and at $\phi=0$ gives $\alpha\rho'(0) + \beta g'(0) = 0$ with $g'(0) = \int_0^1 \rho = 1$, so $\rho'(0) = -\beta/\alpha = -\tau^2$. Substituting $\rho(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi)$ into (13) and (14) together with $\int_0^1 \rho = 1$ gives the unique positive interior stationary density

$$\rho_{\tau}(\phi) = \frac{\tau \cosh(\tau(1 - \phi))}{\sinh \tau}, \quad (15)$$

which is main-text Theorem 4. The mass constraint $\int \rho=1$ is not lost: it enters the boundary condition at $\phi=0$ via $g'(0)=\int_0^1 \rho=1$. Positivity holds on the closed interval since $\rho_{\tau}(\phi) \geq \rho_{\tau}(1) = \tau/\sinh \tau > 0$ for $\tau > 0$ (with $\rho_{\tau}(0) = \tau \coth \tau > 0$); the inequality constraint $\rho > 0$ is therefore inactive throughout $[0, 1]$ and the unconstrained Euler–Lagrange solution coincides with the KKT-constrained solution. For the degenerate case $\beta=0$ the ODE reduces to $\rho''=0$ with $\rho'(0)=\rho'(1)=0$ and $\int \rho=1$, forcing $\rho \equiv 1$, which is also the $\tau \rightarrow 0$ limit of (15).

Convexity / PSD of $\mathcal{C}_{\text{app}}$. For $\alpha > 0$ and $\beta \geq 0$, the surrogate is convex because both quadratic terms are positive semidefinite. The diagonal $\alpha \int \rho^2$ is manifestly PSD on L^2 , and the Green-kernel cross term satisfies the identity

$$\iint_{[0,1]^2} f(\phi) f(\psi) \min(\phi, \psi) d\phi d\psi = \int_0^1 \left(\int_s^1 f(u) du \right)^2 ds \geq 0. \quad (16)$$

Thus the stationary density obtained above is the unique constrained *minimizer* (not merely a critical point), and the surrogate is well-defined as a quadratic functional on $L^2([0, 1]) \supset C^2([0, 1])$ with no further regularity assumption needed for the δ -component. Conditional on $\mathcal{C}_{\text{app}}$, this derivation is exact.

Why this branch. The homogeneous Cosh solution has an analytically invertible CDF, turning the continuous optimum into an explicit finite table. The functional test below checks the resulting allocation against a closed-form cosine-feature slice, and the trained comparisons supply the behavioural test.

A.5 COSINE-SLICE DIAGNOSTIC FOR THE SURROGATE-DERIVED TABLE

The broadband surrogate $K_{\text{app}} = \alpha\delta + \beta \min$ supplies the closed-form density. We validate its allocation functionally against the closed-form cosine-feature kernel $K_{\text{cos}}(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta) d\Delta$: does the derived allocation also reduce the collision objective under K ? This tests whether the same direction holds for the declared cosine-feature slice.

Table 3 answers affirmatively. For a uniform distance prior $D(\Delta) = 1/L$ on $[0, L]$, we compute K_{cos} at the discrete allocation points for Geo and EVQ-COSH, then evaluate the normalized collision score $C = \sum_{i < j} K_{ij}^2 / (K_{ii}K_{jj})$ and Shannon effective rank (the exponential of the cosine-feature Gram eigenvalue distribution). The inclusive grid $u_k = k/(K-1)$ exposes the complete diagnostic interval. This slice-level static diagnostic is separate from the phase-invariant full-subspace geometry of Theorem 1; trained midpoint tables are evaluated in §4. The 12 reported configurations comprise a six-length text sweep at fixed $b=500K$ and $K=32$, two additional text bases at $L=2048$, and four video bases at $L=32$ and $K=16$.

Table 3: Cosine-slice surrogate check on the inclusive $u_k=k/(K-1)$ diagnostic grid. EVQ-COSH reduces collision and increases Shannon effective rank of the cosine-feature Gram under the closed-form kernel K_{cos} in all 12 tested configurations. Theorem 1 supplies the separate phase-invariant full-subspace geometry; §4 evaluates trained midpoint tables.

Configuration	Collision C		Shannon rank	
	Geo	EVQ-COSH (Δ_C)	Geo	EVQ-COSH
<i>Text: $K=32, b=500K, \text{ sweep } L$</i>				
$L=128$	226.3	17.7 (−92%)	2.5	17.0
$L=256$	192.6	27.6 (−86%)	3.0	13.9
$L=512$	163.2	43.3 (−73%)	3.6	10.8
$L=1024$	133.9	59.0 (−56%)	4.4	8.8
$L=2048$	109.0	67.5 (−38%)	5.4	8.0
$L=4096$	87.0	66.3 (−24%)	6.6	8.1
<i>Text: $K=32, L=2048, \text{ sweep } b$</i>				
$b=10K$	34.8	19.2 (−45%)	13.0	17.3
$b=100K$	79.4	47.0 (−41%)	7.1	10.4
<i>Video: $K=16, L=32, \text{ sweep } b$</i>				
$b=100$	22.6	14.2 (−37%)	4.8	6.4
$b=1K$	42.3	28.0 (−34%)	2.8	3.9
$b=10K$	55.4	39.8 (−28%)	2.2	2.9
$b=50K$	63.3	47.2 (−26%)	2.0	2.5

EVQ-COSH reduces collision under K_{cos} in all 12 configurations (24%–92%), and Shannon effective rank increases in every row. In the fixed-base text sweep, the relative reduction grows as L shortens. The separate base sweeps remain positive but are not monotone in b .

A.6 SINGLE-CROSSING BUDGET SHIFT

Lemma 7 (Single-crossing budget shift). *For every $\tau > 0$, ρ_τ crosses the uniform density exactly once, at*

$$\phi_c(\tau) = 1 - \tau^{-1} \operatorname{arcosh}(\sinh \tau / \tau) \leq 1 - 1/\sqrt{3}. \quad (17)$$

Proof of Lemma 7. For $\tau > 0$, $\rho_\tau(\phi) = \tau \cosh(\tau(1 - \phi)) / \sinh \tau$ is strictly decreasing on $[0, 1)$, with $\rho_\tau(0) = \tau \coth \tau > 1$ and $\rho_\tau(1) = \tau / \sinh \tau < 1$. Hence the crossing is unique, and solving $\rho_\tau(\phi_c) = 1$ gives (17). To bound it, expand

$$\frac{\sinh \tau}{\tau} = \sum_{n \geq 0} \frac{\tau^{2n}}{(2n+1)!} \geq \sum_{n \geq 0} \frac{\tau^{2n}}{3^n (2n)!} = \cosh\left(\frac{\tau}{\sqrt{3}}\right),$$

where the coefficient-wise inequality is $3^n \geq 2n + 1$. Monotonicity of arcosh yields $\operatorname{arcosh}(\sinh \tau / \tau) \geq \tau / \sqrt{3}$, proving the bound. The three-region masses follow directly from $F_\tau(\phi) = 1 - \sinh(\tau(1 - \phi)) / \sinh \tau$ at $\phi_L = \log(L/2\pi) / \log b$ and $\phi_M = \log(M/2\pi) / \log b$. $\square$

A.7 SURROGATE SELF-CONSISTENCY THEOREM

The surrogate $\mathcal{C}_{\text{app}}$ contains two integral terms $T_1(\tau) = \int_0^1 \rho^2 d\phi$ and $T_2(\tau) = \int_0^1 \int_0^1 \rho(\phi) \rho(\psi) \min(\phi, \psi) d\phi d\psi$, where $\rho(\phi) = \tau \cosh(\tau(1 - \phi)) / \sinh \tau$ is the normalized cosh density.

Theorem 8 (Surrogate self-consistency). *For all $\tau > 0$:*

$$\tau^2 T_2(\tau) + T_1(\tau) = \tau \coth \tau. \quad (18)$$

Proof. The Green’s function $g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi$ satisfies $g''(\phi) = -\rho(\phi)$ with $g(0) = 0$, $g'(1) = 0$ (from the min kernel structure; see §A). Define $h(\phi) = \rho(0) - \rho(\phi)$. Then $h'' = -\rho''$, and under the pure-tether ODE $\rho'' = \tau^2 \rho$ this gives $h'' = -\tau^2 \rho$, with $h(0) = 0$ and $h'(0) = -\rho'(0)$.

Define $f(\phi) = \tau^2 g(\phi) - h(\phi)$. Then $f'' = \tau^2 g'' - h'' = -\tau^2 \rho - (-\tau^2 \rho) = 0$, so f is linear in ϕ . The boundary values give $f(0) = \tau^2 g(0) - h(0) = 0$ and $f'(0) = \tau^2 g'(0) + \rho'(0) = \tau^2 - \tau^2 = 0$ (using $g'(0) = \int_0^1 \rho = 1$ and $\rho'(0) = -\tau^2$ from the pure-tether density). Hence $f \equiv 0$, yielding $\tau^2 g(\phi) = h(\phi) = \rho(0) - \rho(\phi)$, i.e.,

$$g(\phi) = \frac{\rho(0) - \rho(\phi)}{\tau^2}. \quad (19)$$

Integrating against ρ :

$$T_2 = \int_0^1 \rho(\phi) g(\phi) d\phi = \frac{1}{\tau^2} [\rho(0) \underbrace{\int_0^1 \rho}_{=1} - T_1] = \frac{\rho(0) - T_1}{\tau^2}. \quad (20)$$

Since $\rho(0) = \tau \cosh \tau / \sinh \tau = \tau \coth \tau$, rearranging gives (18). $\square$

An immediate corollary follows.

Corollary 9 (Closed-form T_2).

$$T_2(\tau) = \frac{\sinh 2\tau - 2\tau}{4\tau \sinh^2 \tau}. \quad (21)$$

This follows by substituting $T_1(\tau) = \tau^2(1 + \sinh(2\tau)/(2\tau))/(2\sinh^2 \tau)$ and $\rho(0) = \tau \coth \tau$ into the theorem.

The identity checks the two surrogate energy terms exactly; trained-loss evidence is supplied by §4.

A.8 FROM SCALING STRUCTURE TO AN OPERATING DEFAULT

The implementable reference used in standard full-RoPE text experiments is

$$\tau = c d_{\text{head}} / \sqrt{L_{\text{train}}}, \quad u_k = (k + \tfrac{1}{2}) / K, \quad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}. \quad (22)$$

Here d_{head} is the full-RoPE head width and $c=1$ is fixed before training for the reported zero-search reference; realised τ , grid, and endpoint conventions are recorded with each protocol.

The post-softmax model uses $\mathcal{F}(\tau) = \frac{1}{2}S_{\chi^2}(\tau) - \lambda U(\tau, L)$, where $\lambda > 0$ is a fixed trade-off weight, with $S_{\chi^2}(\tau) = \tau^4/(45d_{\text{head}}) + O(\tau^6)$ and $U(\tau, L) = (d_{\text{head}}/L)[Q_0(L, b) + \tau^2 Q_1(L, b) + O(\tau^4)]$. Its leading stationary balance is

$$\tau_*^2 = 45\lambda Q_1(L, b) \frac{d_{\text{head}}^2}{L}. \quad (23)$$

Thus the model supplies the powers of d_{head} and L ; its local coefficient $c_{\text{loc}}(L, b, \lambda) = \sqrt{45\lambda Q_1(L, b)}$ depends on the chosen utility, base, and trade-off weight. Setting $c=1$ in Eq. (22) is the deployed zero-search convention fixed before training, rather than an estimate fitted to each configuration.

Here

$$q(x) = \frac{1}{2} + \frac{\sin(2x)}{4x} - \left(\frac{\sin x}{x}\right)^2 = \text{Var}_{t \sim U[0,1]}[\cos(xt)]$$

and $Q_0(L, b) = \int_0^1 q(Lb^{-\phi}) d\phi$ and $Q_1(L, b) = \int_0^1 \eta(\phi) q(Lb^{-\phi}) d\phi$. This specified phase-variance utility supplies the scaling model; the exact oscillatory collision objective and trained-transformer loss are evaluated separately.

The finite- τ evidence evaluates a configuration-independent zero-search prior. In a staged 99-run study, the default beats the midpoint-discretised Geo baseline in 7/9 configuration means and 18/27 seed pairs. The evaluated neighbours use the discrete ratios $0.75\times$, $1.25\times$, or $1.5\times$ of the formula; the independent exact-range factorial tests $0.75\times$, $1.00\times$, and $1.25\times$. Their ordering varies across configurations. These tested multipliers sample the formula point and its registered neighbours. The evidence supports $c=1$ as a useful zero-search prior relative to that Geo baseline. The full run matrix and statistical units are recorded in `data/curated/phase16_99run_manifest.csv`; the supplement includes the runner `scripts/core_text_phases/phase16_formula_optimality_sweep.py`. This discrete design establishes the tested operating prior, not a continuous basin or per-configuration optimum.

A.9 POST-SOFTMAX OPERATING-POINT MODEL

The shape surrogate and the operating-point model answer different questions. The surrogate selects the Cosh family; the proposition below gives the scaling structure used to set its finite strength.

Proposition 10 (Semi-analytic operating-point scaling). *On the pure-tether family, assuming $Q_1(L, b) > 0$ and a diffuse post-softmax baseline $p_0=1/L$, the Pearson- χ^2 channel-load stiffness and the post-softmax transport utility admit small- τ expansions $S_{\chi^2}(\tau) = c_S \tau^4/d_{\text{head}} + O(\tau^6)$ with $c_S = \frac{1}{45}$, and $U(\tau, L) = (d_{\text{head}}/L)[Q_0 + \tau^2 Q_1(L, b) + O(\tau^4)]$, where $Q_1(L, b) = \int_0^1 \eta(\phi) q(Lb^{-\phi}) d\phi$, $\eta(\phi) = (1-\phi)^2/2 - 1/6$ and $q(x) = 1/2 + \sin(2x)/(4x) - (\sin x/x)^2$. The leading-order stationary point of $\mathcal{F}(\tau) = \frac{1}{2}S_{\chi^2}(\tau) - \lambda U(\tau, L)$ satisfies*

$$\tau_*^2 = 45\lambda Q_1(L, b) \cdot \frac{d_{\text{head}}^2}{L}, \quad \tau_* \propto \frac{d_{\text{head}}}{\sqrt{L}}.$$

The explicit $d_{\text{head}}/\sqrt{L}$ factor is modulated by $Q_1(L, b)$, so this is a scaling structure rather than a pure power law, and the prefactor $c_{\text{loc}}(L, b, \lambda) = \sqrt{45\lambda Q_1(L, b)}$ remains convention- and protocol-dependent. The proposition supplies a small- τ scaling model; the deployed configurations near $\tau=4$ use the exact Cosh warp. The trained-model controls in App. C evaluate the independently fixed empirical default $c=1$.

A.10 CHANNEL-LOAD MOTIVATION FOR PEARSON χ^2

Normalization convention. Throughout this paper, $S_{\chi^2}(\tau)$ denotes the d_{head} -normalized stiffness $S_{\chi^2}(\tau) := \frac{1}{d_{\text{head}}} \int_0^1 (1-\rho_\tau)^2 / \rho_\tau d\phi$, with leading expansion $\tau^4/(45d_{\text{head}}) + O(\tau^6)$ consistent with Proposition 10. Identities below that produce a bare $\tau^4/45$ leading coefficient apply to $d_{\text{head}} S_{\chi^2}(\tau)$ (the un-normalized integral); the d_{head} factor enters via the surrogate diagonal $\alpha=1/d_{\text{rot}}$ ($=1/d_{\text{head}}$ for MHA via §A.8; $=1/d_{\text{rot}}$ for compressed-RoPE MLA via §F.1).

Channel-load axioms. Inverse-CDF quantization places channels with cell width $\ell(\phi)=1/\rho(\phi)$ and samples channels according to $dP_\rho=\rho d\phi$. Under the modeling axioms (i) channel-measure averaging, (ii) load ℓ as the primitive local quantity, (iii) second central moment, (iv) uniform density has zero stiffness, (v) local additivity, the canonical load-variance penalty is

$$S(\rho) = \text{Var}_{\phi \sim \rho d\phi} \left[\frac{1}{\rho(\phi)} \right] = \int_0^1 \frac{(1 - \rho(\phi))^2}{\rho(\phi)} d\phi = S_{\chi^2}(\rho). \quad (24)$$

Under the stated axioms, this is the canonical load-variance penalty.

A.1.1 DISCRETE-CHANNEL TRANSPORT GAP UNDER FINITE CHANNEL COUNTS

Setup. Inverse-CDF quantization replaces the continuous density ρ by an atomic measure. Let $F(\phi)=\int_0^\phi \rho$, $Q(u)=F^{-1}(u)$, midpoint grid $u_k=(k-\frac{1}{2})/K$, and $\phi_k=Q(u_k)$. The empirical channel measure is $\mu_K=K^{-1}\sum_{k=1}^K \delta_{\phi_k}$. Direct $\|\rho_K - \rho\|$ is ill-posed: one cannot subtract an atomic measure from a density. We therefore work in Wasserstein distance for transport and in a quantile-cell histogram ρ_K for density.

Transport bounds. Assume $0 < m \leq \rho \leq M$ and $\|\rho'\|_\infty \leq B$; then Q is $(1/m)$ -Lipschitz. The midpoint quantization errors satisfy

$$W_\infty(\mu_K, \mu_\rho) \leq \frac{1}{2Km}, \quad W_1(\mu_K, \mu_\rho) \leq \frac{1}{4Km}. \quad (25)$$

Defining the quantile-cell histogram $\rho_K(\phi)=1/(K|I_k|)$ on $I_k=[Q((k-1)/K), Q(k/K)]$, the histogram density satisfies

$$\|\rho_K - \rho\|_1 \leq \frac{B}{Km}, \quad \|\rho_K - \rho\|_\infty \leq \frac{B}{Km}. \quad (26)$$

The transport bounds couple each quantile cell to its midpoint; the histogram bound follows from the mean-value theorem and $|I_k| \leq 1/(Km)$.

Application to EVQ-Cosh. For $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi))/\sinh \tau$, $m_\tau = \tau/\sinh \tau$ and $B_\tau = \tau^2$. Substituting,

$$W_1(\mu_{K,\tau}, \mu_{\rho_\tau}) \leq \frac{\sinh \tau}{4K\tau}, \quad \|\rho_{K,\tau} - \rho_\tau\|_1 \leq \frac{\tau \sinh \tau}{K} = \frac{\tau^2}{K} + O\left(\frac{\tau^4}{K}\right). \quad (27)$$

Kernel-integrated errors. For a smooth two-variable kernel K_{sm} with Lipschitz constant L_K in each argument,

$$\left| \iint K_{\text{sm}} d\mu_K d\mu_K - \iint K_{\text{sm}} d\mu_\rho d\mu_\rho \right| \leq \frac{L_K}{2Km}. \quad (28)$$

The kernel bound applies the W_1 estimate to each argument. The singular δ -component of K_{app} is represented by its cell average.

High-resolution distortion. For midpoint scalar quantisation with a smooth weighted distortion w , the standard high-resolution expansion is

$$\mathcal{D}_K[\rho] = \frac{1}{12K^2} \int_0^1 \frac{w(\phi)}{\rho(\phi)^2} d\phi + O(K^{-3}). \quad (29)$$

This is the finite-channel mechanism: a shaped ρ can reduce the weighted inverse-density load $\int w/\rho^2$; the absolute benefit scales as K^{-2} for smooth quantization error and as K^{-1} for transport/kernel terms.

Dependence of the bounds on channel count. The standard MHA configuration has $d_{\text{head}}=d_{\text{rope}}=64$ and $K=32$; the MLA study has $d_{\text{head}}=64$, $d_{\text{rope}}=32$, $d_{\text{nope}}=32$ and $K=16$. Halving K scales the K^{-2} component of this distortion bound by $4\times$ and the K^{-1} component by $2\times$. The MLA experiment (§F.1) supplies the corresponding trained-model test at $K=16$.

Large- τ regime. At large τ , $m_\tau = \tau / \sinh \tau$ becomes small and the $1/(Km_\tau)$ factor grows. The transport bounds are informative while this factor remains controlled.

B EXPERIMENTAL DETAILS

B.1 SCOPE OF THIS APPENDIX

The exact-range and mature-model protocols are documented separately in Apps. C and E. This section records the scratch-model and cross-modal experiments that remain supporting evidence. The learned-frequency comparator appears in App. D.

B.2 REPRODUCIBILITY SNAPSHOT

The evidence ladder separates causal identification from architecture, scale, adaptation, frozen-checkpoint, and cross-modal consequences. Scratch text runs use FineWeb-Edu (Penedo et al., 2024).

Table 4: Evidence and protocol map. Every row retains its own intervention, experimental unit, and endpoint.

Role	Model/regime	Intervention	Unit	Endpoint	Appendix
Allocation identification	151.9M scratch	fixed support, 30 interiors	3 training seeds	tail NLL	C.1
Shape/configuration breadth	50.9M factorial	fixed support, 5 allocations	12 configs $\times$ 3 seeds	weighted OOD NLL	C.2
Scarce rotary budget	432M MLA-style	Geo/EVQ-COSH, $K=16$	3 training seeds	PPL	F.1
Full-parameter scale	750M/1.485B	continued/from-init Geo/EVQ-COSH	one matched pair each	PPL/retrieval	B.3/E.1
Frozen allocation	OLMo/Qwen	same support and amplitude	20 rows/task	RULER	E.2
Adaptation/source use	1.485B/8B	matched LoRA/source block	one trained pair	QA/RULER/NLL	E.1/E.3
Cross-modal scope	129.6M video DiT	temporal Geo/EVQ-COSH	1 training seed	denoising MSE	B.5

Table 5 lists the training hyperparameters shared across the supporting text experiments.

Table 5: Training hyperparameters for the supporting scratch text experiments.

Parameter	Value
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$, $\epsilon=10^{-8}$)
Learning rate	6×10^{-4} (50M), 2×10^{-4} (MLA/750M)
LR schedule	Cosine decay to $0.1 \times$ peak LR
Warmup	2% of total steps (linear)
Batch size (seqs)	32 (50M), 2–4 (750M), 6 (MLA)
Weight decay	0.1
Gradient clipping	max-norm 1.0
Dropout	0.0
Sequence packing	Contiguous, no cross-document masking
<i>Video DiT (129.6M)</i>	
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$)
RoPE base	10,000 (temporal axis)
Activation	GELU (tanh approx.)
<i>MLA (432M)</i>	
LR / warmup	2×10^{-4} / 2% of steps
Training tokens	500M

Within each text comparison, the architecture, optimizer, data order, and training loop are fixed, and the only intervention is the initialized inverse-frequency table. The scratch-text experiments use base $b=500K$ unless stated otherwise.

B.3 LARGER-SCALE SUPPORTING EVIDENCE

The 750M ‘continue@4K’ experiment starts both arms from the same 2K Geo checkpoint and applies a 500M-token continuation with either the Geo or EVQ-COSH table (seed 42). At 16K, PPL is

Table 6: 750M continued-pretraining check (seed 42; 2K $\rightarrow$ 4K; 500M continuation tokens). Passkey retrieval is teacher-forced NLL-gap and is saturated at 8K; AR exact is greedy autoregressive exact match over 40 trials.

Metric	Geo	EVQ-COSH
PPL@2K	25.9	26.2
PPL@4K	22.0	22.3
PPL@8K	23.4	19.6
PPL@16K	45.1	24.4
NLL-gap retrieval @8K	100%	100%
Strict AR exact @8K	0%	77.5%
Global strict AR exact	66.67%	92.5%

45.1 versus 24.4; on 40 passkey trials at 8K, strict autoregressive exact match is 0% versus 77.5%. The EVQ-COSH continuation uses midpoint quantisation with $\tau=1.5$. Both arms are full-parameter continuations from the shared checkpoint.

B.4 MODALITY-DEPENDENT τ CORRECTION FOR VIDEO DiT

The text-style reference used for this sweep gives $\tau=K_t/\sqrt{T}=2.83$ for $K_t=16$ and $T=32$; the best tested Video DiT point is $\tau=1.5$ (0.53 $\times$), which selects the supporting video configuration.

B.5 CROSS-MODAL EVIDENCE: VIDEO DiT WITH BIDIRECTIONAL ATTENTION

VideoRoPE and RIFLEx make temporal-frequency allocation explicit in video models (Wei et al., 2025; Zhao et al., 2025). We test the same allocation principle in a 129.6M video diffusion transformer with bidirectional attention and 3D RoPE on Oscillating Moving MNIST (Srivastava et al., 2015). Training uses 32 frames; evaluation uses 128 frames (4 $\times$) under the same fixed temporal range transform in both arms. The seed-42 Geo and EVQ-COSH ($\tau=1.5$) branches are trained in the same head-to-head run.

Table 7 reports the matched comparison. EVQ-COSH reduces denoising MSE by 21% on training frames, 16% over all extrapolated frames, and 35% on far extrapolated frames, extending the scope check to bidirectional 3D RoPE.

Table 7: DiT head-to-head evaluation (129.6M, seed 42, bidirectional attention, 3D RoPE). Values are denoising MSE at $t=0.5$.

Seed	Metric	Geo ($\tau=0$)	EVQ-COSH ($\tau=1.5$)	Δ
42	Train MSE	0.00911	0.00720	−21%
42	All extrapol	0.00724	0.00606	−16%
42	Far extrapol	0.00989	0.00639	−35%

Low-phase-channel diagnostic. At base = 10,000, six of the 16 temporal frequencies accumulate less than 0.1 rad over the 32 training frames. This analytic count identifies the low-phase regime tested by the matched behavioural comparison above.

B.6 WHAT THE SCHEDULE CONCRETELY DOES

At $\tau=4$, $b=5 \times 10^5$, the warp moves budget toward short inverse-frequency scales: the continuous quantile $u=0.5$ moves from $1/\omega=707$ to 9.69, and $u=0.9$ from 134,609 to 1,720. The deployed midpoint grid therefore changes sampled extrema and span at finite K ; the exact-range experiments anchor those quantities explicitly. At $\tau=2$, $\rho_\tau(0.1)=1.71$ and $\rho_\tau(1)=0.55$, while the band $1/\omega=2048$ has relative density 0.76. Thus the finite grid thins most strongly at scales much slower than the training window while retaining more of the training-scale band. The exact-range experiments

measure the corresponding trained allocation effect, and Table 1 separately measures incompatibility after a frozen post-hoc swap.

C IDENTIFICATION PROTOCOL AND PROVENANCE

C.1 EXACT-RANGE CONTROL AT 151.9M

What is pinned. The control is designed so that every scalar quantity a base/range method can express is identical between arms. Within each seed pair the two arms share: model architecture (151,898,880 parameters); trainable initialisation; training length 256; token order; optimiser, LR schedule, global batch; within-pair micro batch and accumulation; training budget (499,974,144 tokens, 7,629 optimiser steps); the 32 frozen evaluation anchors; the sampled highest frequency; the sampled lowest frequency; and the log-frequency span. The single free variable is the location of the $K-2 = 30$ interior frequencies.

The optimiser is fused AdamW with $\beta_1=0.9$, $\beta_2=0.95$, weight decay 0.01, and gradient clipping at 1.0. Peak/minimum learning rates are $6 \times 10^{-4}/6 \times 10^{-5}$ with 762 warmup steps and cosine decay. The global batch is 256 sequences of length 256 for every arm. Seed 42 uses micro batch 64 with four accumulation steps; seeds 137 and 256 use micro batch 128 with two accumulation steps. Each paired comparison therefore has identical execution geometry, optimiser steps, and counted-token budget; the aggregate uses training seed as its statistical unit under this shared scientific protocol.

Model and data. The model has 12 layers, hidden size 768, 12 heads of dimension 64, MLP width 3,072, and vocabulary size 50,304. Training uses the unshuffled FineWeb-Edu (Penedo et al., 2024) sample/10BT/000_00000.parquet shard at pinned revision 87f09149ef4734204d70ed1d046ddc9ca3f2b8f9; validation uses the disjoint sample/10BT/004_00000.parquet shard. Both are tokenised with the GPT-NeoX-20B tokenizer (Black et al., 2022) at revision c292233c833e336628618a88a648727eb3dff0a7.

Arms. *Baseline:* FMRoPE at fixed training support, implementing the uniform-in-log exponent grid specified in §6.1 of Oka et al. (2026b) at training base 256. *Intervention:* anchored EVQ-COSH at $\tau=4$. The anchored and deployed midpoint EVQ-COSH tables are different support embeddings of the *same normalised shape*: both use midpoint Cosh quantiles q_k , and the diagnostic removes only their affine endpoint degrees of freedom. This normalisation is what makes the exact-range control possible.

Concretely, let $q_k = \Phi_\tau((k + \frac{1}{2})/K)$, $s_k = (q_k - q_0)/(q_{K-1} - q_0)$, and $R = ((K-1)/K) \log B$. The intervention uses

$$\omega_k^{\text{anchor}} = \exp(-Rs_k), \quad \omega_k^{\text{FMR}} = \exp\left(-R\frac{k}{K-1}\right). \quad (30)$$

Hence $k=0, K-1$ match exactly and only the $K-2$ interior entries differ.

Evaluation. Two conditions. *fixed:* each checkpoint retains its training range. *target-matched:* the range is set to the declared evaluation length while preserving each arm’s normalised interior spacing. Metric is paired final-128-token teacher-forced NLL over the 32 frozen anchors; reported PPL is $\exp(\text{mean NLL})$.

Three-seed contrasts. Negative values favour anchored EVQ-COSH. The aggregation unit is the training seed; evaluation anchors remain paired observations within a seed.

Under *fixed*, anchored EVQ-COSH is favoured by 3/3 seeds at every OOD length; under *target-matched*, FMRoPE is favoured by 3/3 seeds at every OOD length. The fixed-range sign agrees across all seeds and OOD lengths; the 1K/2K magnitudes are the most stable across seeds.

Reading. The *fixed* column identifies the optimisation variable: interior allocation is not reducible to scalar base/range selection. Under the tested target-retargeted policy, the ordering of the two tested policies reverses. The two columns therefore establish that the fixed-support result is conditional on the selected range; they do not identify a support-by-allocation interaction law. The *fixed* column isolates allocation when one table serves multiple lengths (§2.1).

Table 8: Exact-range anchored EVQ-COSH minus FMRoPE tail-NLL contrasts. Mean $\pm$ SD is across three independent training seeds.

Length	Fixed training range				Target-matched range			
	Seed 42	Seed 137	Seed 256	Mean $\pm$ SD	Seed 42	Seed 137	Seed 256	Mean $\pm$ SD
256	+0.033	+0.029	+0.017	+0.026 $\pm$ 0.008	+0.033	+0.029	+0.017	+0.026 $\pm$ 0.008
512	-0.478	-0.271	-0.094	-0.281 $\pm$ 0.192	+0.061	+0.053	+0.067	+0.060 $\pm$ 0.007
1,024	-0.205	-0.191	-0.132	-0.176 $\pm$ 0.039	+0.182	+0.156	+0.344	+0.227 $\pm$ 0.102
2,048	-0.113	-0.181	-0.143	-0.146 $\pm$ 0.034	+0.279	+0.399	+0.701	+0.460 $\pm$ 0.218

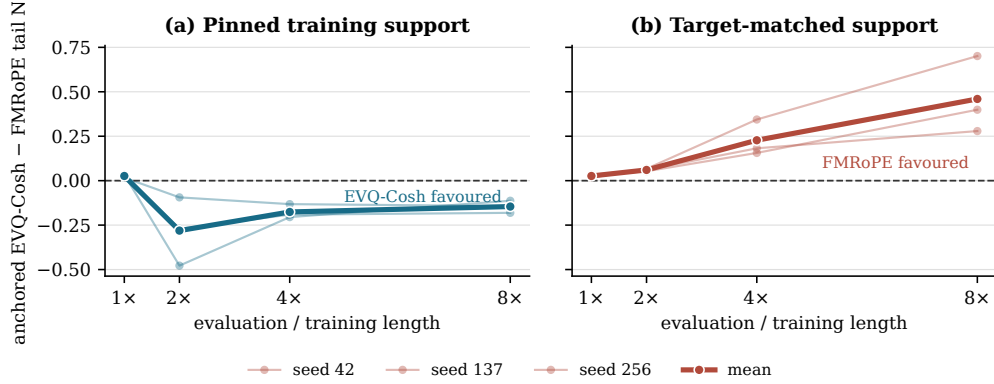


Figure 4: **Support and interior allocation are distinct design coordinates.** Each thin line is one training seed and the heavy line is the mean. At pinned training support, all three seeds favour anchored EVQ-COSH at every OOD length (a). Retargeting each arm’s support reverses all three OOD comparisons (b). The fixed-support panel isolates the allocation effect; the tested target-matched policy reverses the ordering of the two tested policies.

C.2 EXACT-RANGE FACTORIAL AT 50.9M

Design. The schedule arms, including the $1.25\times$ multiplier, were specified before any result was inspected. Grid: 2 bases $\times$ 2 training lengths $\times$ 3 head dimensions = 12 structural configurations; 5 schedules (Geo, anchored EVQ-COSH at $0.75\times$, $1.00\times$ [the formula point], $1.25\times$, and a deformation-matched exponential); 3 seeds. Main factorial 180/180 completed, boundary- τ follow-up 12/12 completed, covering the complete pre-specified matrix.

The exponential control uses $z_\gamma(t) = (e^{\gamma t} - 1)/(e^\gamma - 1)$ on $t \in [0, 1]$. Its γ is chosen without task results so that the RMS node displacement from the uniform grid matches the formula-Cosh displacement, while the sampled endpoints and log-span remain identical across arms.

Metric. Weighted OOD NLL over $2L/4L/8L$,

$$\bar{N}_{\text{OOD}} = \frac{\sum_{r \in \{2,4,8\}} \log_2(r+1) N_{rL}}{\sum_{r \in \{2,4,8\}} \log_2(r+1)},$$

paired within structural configuration and seed, averaged within seed, then equally weighted over the 12 configurations. The statistical unit is the configuration. Equivalent-PPL figures quoted anywhere are $\exp(\text{mean NLL})$, never an arithmetic mean of raw PPL.

Absolute values. Mean weighted OOD NLL (equivalent PPL): Geo 6.211973 (498.68); EVQ-COSH $0.75\times$ 6.202858 (494.16); EVQ-COSH $1.00\times$ 6.202094 (493.78); EVQ-COSH $1.25\times$ 6.199873 (492.69); matched exponential 6.201354 (493.42).

Boundary- τ arms. At formula $\tau=1$ the $1.5\times$ arm is better; at formula $\tau=8$ the $0.75\times$ arm is. The improving direction points inward at both tested extremes; these discrete arms do not locate continuous basin boundaries or yield a replacement closed-form rule.

Table 9: **Exact-range factorial** (50.9M; 2 bases $\times$ 2 training lengths $\times$ 3 head dimensions = 12 configurations, 3 seeds). All arms share sampled extrema and log-span with Geo. Metric: weighted OOD NLL over $2 \times /4 \times /8 \times$, paired within configuration and seed. All arms were pre-specified. CIs are 10,000 configuration-level bootstraps; p is a two-sided exact sign-flip over all 2^{12} assignments, unadjusted across contrasts, so the claim rests on the cross-configuration direction (App. C).

Contrast	Δ NLL	95% CI	$\Delta < 0$ configs	p
EVQ-COSH $0.75 \times$ – Geo	−0.00912	[−0.018, −0.001]	8/12	0.082
EVQ-COSH rule – Geo	−0.00988	[−0.021, 0.002]	7/12	0.125
EVQ-COSH $1.25 \times$ – Geo	−0.01210	[−0.021, −0.003]	10/12	0.027
Matched exponential – Geo	−0.01062	[−0.021, −0.001]	9/12	0.071
<i>The matched exponential reproduces the non-uniform direction:</i>				
EVQ-COSH rule – exponential	+0.00074	[−0.006, 0.008]	7/12	0.836

Scope. This is a short-budget 50.9M mechanistic NLL control trained on a deterministically repeated WikiText-2 raw stream (Merity et al., 2017), a different corpus from the 151.9M control and from the mature-scale runs. The factorial establishes cross-configuration *direction*, while the 151.9M control provides the effect size within its configuration. Scale is evaluated in §4.3.

D LEARNED-FREQUENCY COMPARATOR

Table 10: Learned-frequency comparator (125M, $L_{\text{train}}=128$, FineWeb-Edu, $128 \rightarrow 8\text{K}$). Geo, learned-inv-freq, and EVQ-COSH are seed 42; learnable- τ is 3-seed (42/137/256). Allocation-shape attribution uses the fixed, zero-parameter analytic schedules of §2.1; this separately reported arm is a layer-shared learnable inverse-frequency baseline — one learned scalar per band, shared across layers and heads — and therefore belongs to the learned-*table* family (Karypis et al., 2026); DAPE belongs to the learned positional-*operator* family (Zheng et al., 2024).

Method (128 $\rightarrow$ 8K)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	—
Learnable τ	1	181.2 $\pm$ 1.3	437.9 $\pm$ 12.2	−14.8%
Learned inv-freq (100 $\times$ positional LR)	32	183.6	455.3	−11.4%
same, 10 $\times$ positional LR	32	—	477.7	−7.0%
EVQ-COSH	0	182.0	333.7	−35.0%

The 32-parameter arm in Table 10 is a layer-shared learnable inverse-frequency baseline: one learned scalar per frequency band, shared across layers and heads. It received a dedicated positional-parameter learning-rate sweep at 10 $\times$ and 100 $\times$ the base rate, and the stronger 100 $\times$ setting is the 455.3 row. It answers the learned-table tuning question. Allocation-shape attribution uses the fixed analytic schedules under an identical zero-parameter operator in §2.1.

On method identity: DAPE (Zheng et al., 2024) is a learned positional *operator*, whereas this comparator learns inverse frequencies directly and belongs to the learned-*table* family later studied at scale by Karypis et al. (2026). In this protocol, the learned table improves over Geo but remains weaker than EVQ-COSH; the comparison does not identify whether the gap comes from the final table, optimization dynamics, or the positional-parameter learning rate.

E MATURE-SCALE PROTOCOLS

E.1 OLMo-2 1.485B

All OLMo-2 EVQ-COSH arms use the endpoint grid $u_k=k/K$ with $\tau=2$ and base 500,000; the frequency tensor is fixed throughout each continuation.

Matched routing conversion. We use OLMo-2-0425-1B-Instruct (OLMo Team, 2025) (1.485B actual parameters) with rank-64, alpha-128 Q/K/V/O LoRA (Hu et al., 2022). Native and EVQ-COSH

Table 11: **RULER 13-family scores (%)**. Within each protocol, continuation data and evaluation are matched and one seed is used per arm; evaluation uses 20 examples per family and length. OLMo Q/K-only freezes inherited V/O LoRA and trains only the tensors RoPE acts on; LLaMA uses matched Q/K/V/O LoRA. Training and evaluation rows are disjoint and share the 13 generator families, measuring task-family-adapted length transfer.

Model / protocol	Arm / metric	1 ×	2 ×	4 ×
OLMo-2 1.485B, full Q/K/V/O (4K cap)	Native, official-scorer macro	82.16	0.08	0
	EVQ-COSH, official-scorer macro	37.51	21.29	6.13
OLMo-2 1.485B, Q/K-only (4K cap)	Native, official-scorer macro	72.19	2.02	0.38
	EVQ-COSH, official-scorer macro	42.44	31.63	5.03
LLaMA-3-8B, Q/K/V/O (8K cap)	Native, official-scorer macro	94.44	0.295	—
	EVQ-COSH, official-scorer macro	77.60	14.03	—
	Native, normalized exact	17.69	0	—
	EVQ-COSH, normalized exact	21.54	1.54	—

share the checkpoint, training rows and order, token budget, optimizer, and evaluation rows; the fixed frequency substrate is the intended intervention. Every backward pass contains at most 4K physical tokens, but explicit position IDs expose target-range relative phases. This is therefore a physical-memory cap, not phase-free 4K training. On 8K `niah_single_1`, strict first-generated-number exact is 0/100 for Native and 69/100 for the matched EVQ-COSH seed; an independently trained EVQ-COSH seed scores 67/100 on the same rows. The two EVQ-COSH results are training-seed replications on a shared evaluation set. Natural-text NLL and the distinct autoregressive endpoints are kept separate in the discussion below and in Table 11.

Selective Q/K phase adaptation. To test a less intrusive mature-checkpoint adaptation, we start from the protocol-matched Native and EVQ-COSH Stage-A parents, freeze every inherited V/O LoRA tensor within each arm (verified bitwise), and continue only the existing Q/K tensors for 300 steps. Each arm trains 8,388,608 Q/K parameters while freezing the 8,388,608 inherited V/O parameters, with global batch 8, learning rate 5×10^{-5} , and 20 warmup steps. The QA and RULER generators use seeds 20260728 and 20260729, respectively; the bundled Q/K `metrics.json` records the full protocol. Both arms use the same 1:1:2 mixture of continuous-4K, target-8K, and target-16K phase exposure, complete answer-plus-EOS supervision, and physical sequences no longer than 4K; long phases again use explicit position IDs. On 200 held-out 2WikiMultiHopQA rows per length (Ho et al., 2020), Native/EVQ-COSH token-F1 is 25.99/24.84% at 4K, 0.07/21.48% at 8K, and 0/8.57% at 16K; exact match is 22.0/21.5%, 0/17.5%, and 0/4.0%, respectively. Prompts retain the official question and context and are deterministically filled with answer-filtered 2Wiki distractors to reach the target length, forming a held-out 2Wiki length-transfer evaluation. The RULER adapters are independent of the QA adapters: Table 11 reports both the earlier 276-step full-Q/K/V/O continuation and the 300-step selective-Q/K continuation, each trained over all 13 families. Evaluation uses 20 examples per family and length. A paired family-stratified bootstrap over the tracked Q/K-only per-example predictions gives EVQ-COSH-minus-Native official-scorer macro differences of -29.75 points at 4K (95% CI $[-34.42, -24.85]$), $+29.61$ at 8K ($[26.54, 32.74]$), and $+4.65$ at 16K ($[3.17, 6.38]$). These intervals condition on the single trained pair and quantify evaluation-row uncertainty, not training-seed variability.

From initialisation. Starting from the public step-0 initialisation, the same-recipe Geo/EVQ-COSH comparison gives PPL 177.99/191.36, 161.19/167.45, 163.88/156.87, and 182.73/159.64 at 2/4/8/16K under a matched 2.097B-token budget. The EVQ-COSH-minus-Geo NLL is $+0.0724/+0.0381/-0.0437/-0.1351$ at 2/4/8/16K, with 122/128 and 126/128 held-out documents favouring EVQ-COSH at 8K and 16K. This is one same-initialisation, same-scientific-recipe natural-text LM comparison; capability is evaluated by the separate autoregressive protocols above.

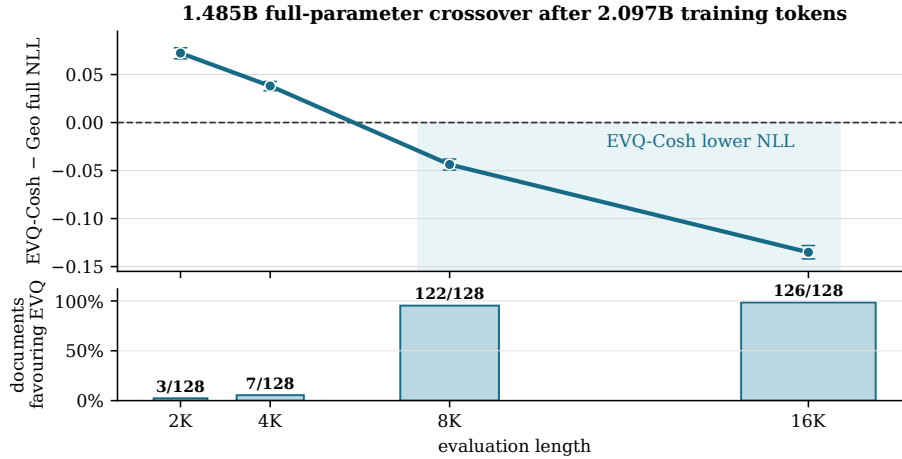


Figure 5: **Full-parameter scale evidence crosses beyond the training window.** The 1.485B Geo and EVQ-COSH arms start from the same public initialisation and follow the same scientific recipe for 2.097B tokens. Error bars are paired bootstrap intervals over the same 128 held-out documents, conditioned on this trained pair; the lower panel gives the document count favouring EVQ-COSH at each length.

E.2 FROZEN FIXED-SUPPORT ALLOCATION

The mature-checkpoint intervention in Table 2 holds both models frozen. The three matched-support arms contain 64 rotary pairs and share the Native fast endpoint and the Native slow endpoint divided by four. The two reference rows do not: Native retains the pretrained table, while official Transformers YaRN applies its frequency-dependent transport. The three matched-support arms also share the factor-four attention amplitude $1 + 0.1 \ln 4$, evaluation rows, greedy decoder, precision, and hardware. The uniform-allocation arm is log-linear between the endpoints. The coarse control is the linear movement ramp with minimum displacement-space MSE to the frozen derived table over the registered discrete profiles; its choice uses no task labels. The remaining arm installs the frozen derived interior allocation.

OLMo-2-0425-1B-Instruct uses nine confirmation-only RULER tasks at 16K, 20 rows per task. Qwen2.5-1.5B-Instruct (Yang et al., 2024) uses the four development RULER tasks at 64K, also with 20 rows per task; it evaluates a factor-four table at twice the Native context length. Row resampling conditions on each fixed checkpoint and task set. Both reallocated tables produce the large OLMo recovery. The derived-minus-uniform-allocation intervals are [54.88, 64.80] points for OLMo and [0.25, 17.50] points for Qwen; the paired derived-minus-coarse-control intervals contain zero on both models. The control therefore attributes the effect to fixed-support reallocation while leaving profile-specific ordering unresolved.

The complete OLMo deployment policy is a separate, bundled intervention: it uses exact Native when observed prefill plus generation reserve fits the Native window and otherwise chooses one frozen factor-four table and amplitude before prefill, without changing the route during the KV-cache lifetime. It adds no learned parameters or training tokens. On a disjoint 128-document FineWeb-Edu evaluation, forcing the same long profile at 4K gives final-1024 NLL 2.8774 versus Native 2.7538, a difference of +0.1236 NLL; the session policy routes 4K requests through Native. At 8/16K, its PG-19 tail NLL is 3.1060/3.0977, compared with 3.4404/3.7955 for official Transformers YaRN factor four. On full-200 Qasper (Dasigi et al., 2021) and 2Wiki (Ho et al., 2020), the routed policy’s token F1 is 24.57/26.66%, versus 18.03/25.69% for YaRN; applying the long profile to all 200 2Wiki rows scores 27.74%. Across the six-task natural-context matrix, the routed method scores 28.58/24.97% at $2/4\times$, compared with 21.27/25.59% for YaRN. The QA arms use identical rows, and every post-chat input plus its generation reserve fits within 16K. These rows evaluate the complete deployment policy on natural context; Table 2 provides the allocation-only contrast.

E.3 LLAMA-3-8B-INSTRUCT

The EVQ-COSH arm in both LLaMA protocols uses midpoint quantisation with $\tau=1.414$; Native retains the pretrained model’s original frequency table. The training objectives, adapters, and endpoints remain separate below.

Probability and causal source use. In the matched seed-42, 300-step Q/K/V/O LoRA protocol, EVQ-COSH-minus-Native NLL is $+0.390/ - 1.510/ - 2.048$ at 8/16/32K. On ten frozen true-16K passkey cases, median target-block hit@16 over 32 retrieval heads rises from 18.75% to 64.06%. Blocking answer-side decode attention to the remote gold block in every head preserves all prompt tokens, original rotary KV indices, and the 128-token block geometry; it changes NLL by -0.0095 for Native and $+1.5055$ for EVQ-COSH. These are matched teacher-forced probability and position-preserving causal-routing measurements; autoregressive task accuracy is evaluated separately. Descriptive Native/EVQ-COSH PPL is 6.817/10.068, 108.958/24.068, and 991.475/127.911 at those lengths.

RULER-family continuation. A separate 516-step continuation uses identical physical-8K supervision over all 13 RULER families, with one seed per arm and disjoint generated training/evaluation rows. Evaluation uses 20 examples per family and length. At 16K, Native/EVQ-COSH official-scorer macro is 0.295/14.03% and normalized exact is 0/1.54%. At 8K, Table 11 reports official scorer macro (94.44/77.60%) and normalized exact (17.69/21.54%).

Endpoint scope. NLL, attention routing, strict autoregressive exact match, multi-hop QA, and RULER remain separate endpoints. The 2Wiki and RULER continuations use the same task families at train and test with disjoint rows, establishing task-family-adapted length transfer. The natural-text and routing rows evaluate probability and causal source use.

F SUPPORTING RESULTS

F.1 MULTI-HEAD LATENT ATTENTION

We test the finite-budget hypothesis in a 432M-parameter transformer following the MLA architectural pattern (DeepSeek-AI, 2024), with 24 layers, width 1024, 16 heads, and per-head width 64. Its decoupled rotary subspace has $d_{\text{rope}}=32$ ($K=16$ frequencies); the remaining $d_{\text{nope}}=32$ dimensions are non-rotary, and the shared KV latent rank is 256. The model trains from scratch on 500M FineWeb-Edu tokens (Penedo et al., 2024) at $L_{\text{train}}=8192$ and base 500K over seeds 42, 43, and 88. EVQ-COSH uses the pre-specified empirical $\tau=1.414$ setting. Because the compressed rotary subspace does not instantiate the full-RoPE $d_{\text{rot}}=d_{\text{head}}$ convention in Eq. (22), this experiment tests frequency allocation at fixed architecture rather than the operating rule.

For the supporting overlay, let $\lambda_k = 2\pi/\omega_k$ and $t_k = \text{clip}((\lambda_k - L_{\text{train}})/(31L_{\text{train}}), 0, 1)$. The MLA wavelength-blend operator sets $\omega'_k = \omega_k/[1 + t_k(s - 1)]$: it preserves wavelengths below the training length, applies the full factor s above $32L_{\text{train}}$, and interpolates between them.

Table 12: MLA validation (432M, $d_{\text{rope}}=32$, 3-seed mean $\pm$ std). Rows marked “MLA wavelength-blend” use this run’s wavelength-blend operator defined above.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
Geo	35.4 $\pm$ 0.9	138.8 $\pm$ 5.5	241.5 $\pm$ 2.6	323.7 $\pm$ 5.2
Geo + MLA wavelength-blend ($s = 4$)	35.5 $\pm$ 0.9	117.9 $\pm$ 6.5	204.9 $\pm$ 6.9	278.5 $\pm$ 6.3
EVQ-COSH	35.8 $\pm$ 0.8	95.6$\pm$4.1	204.9 $\pm$ 14.7	291.6 $\pm$ 20.5
EVQ-COSH + MLA wavelength-blend ($s = 4$)	35.8 $\pm$ 0.8	71.1$\pm$4.1	153.2$\pm$12.5	236.6$\pm$15.9

EVQ-COSH reduces 16K PPL by 31.1% (138.8 $\rightarrow$ 95.6), while 8K PPL changes by $+0.9\%$. All three training seeds favour EVQ-COSH at 16K. The EVQ-COSH plus MLA wavelength-blend row has the lowest mean PPL at 16K and beyond. These rows establish the tested ordering for this architecture and protocol.